Astrodynamics 2

HW 5 & HW 6

2020.12.08

김나은

HW 5

HW 5

- Use of Lambert Problem for Targeting using only Two-Body dynamics

$$\left(\vec{r}_{int}, \vec{r}_{tgt}, \vec{V}_{int}, \vec{V}_{tgt}, \Delta t \Rightarrow \Delta\vec{V}_a, \Delta\vec{V}_b\right)$$

I. Propagate Target through the desired TOF
Kepler $\left(\vec{r}_{tgt}, \vec{V}_{tgt}, \Delta t \Rightarrow \vec{r}_{tgt,b}, \vec{V}_{tgt,b}\right)$

II. Determine Transfer Orbit using Lambert Problem Approach
Lambert-Gauss $(\vec{r}_{\text{int}}, \vec{r}_{tgt,b}, \Delta t \Rightarrow \vec{V}_{trans,a}, \vec{V}_{trans,b})$

III. $\Delta\vec{v}_a = \vec{v}_{trans,a} - \vec{v}_{\text{int}}$
$\Delta\vec{v}_b = \vec{v}_{tgt} - \vec{v}_{trans,b}$

IV. Check for the Earth impact
Hit Earth $(\vec{r}_{\text{int}}, \vec{v}_{trans,a}, \vec{r}_{tgt,b}, \vec{v}_{trans,b}, a \Rightarrow r_p, collision)$

The initial conditions for moving target

Interceptor: r_{int}= 5328.7862 I + 4436.1273 J + 101.4720 K km
v_{int}= -4.864779 I +5.816486 J + 0.240163 K km/sec

Target: r_{tgt}= 6697.4756 I + 1794.5831 J + 0.0000 K km
v_{tgt}= -1.962372 I + 7.323674 J + 0.000000 K km/sec

Time of flight is 100 minutes

HW 5

- Propagate target through the desired TOF
  - Setting initial condition

```
r_int=[5328.7862 4436.1273 101.4720];
v_int=[-4.864779 5.816486 0.240163];

r_tgt=[6697.4756 1794.5831 0];
v_tgt=[-1.962372 7.323674 0];
t=100*60;
mu=398600;
[p,a,e,i,w,argp,nu,m,arglat,truelon,lonper ] = rv2coe (r_tgt,v_tgt, mu);
```

  - Propagate through TOF

```
[r_TGT,v_TGT,errork] =  kepler(r_tgt,v_tgt,t);

[p,a,e,i,w,argp,nu1,m,arglat,truelon,lonper ] = rv2coe (r_TGT,v_TGT, mu);
```

HW 5

- Determine Transfer orbit using Lambert problem approach (Lambert-Gauss)
  - Find l, m

$$l \equiv \frac{r_o + r}{4\sqrt{r_o r}\cos\left(\frac{\Delta\nu}{2}\right)} - \frac{1}{2} \qquad m \equiv \frac{\mu \Delta t^2}{\left\{2\sqrt{r_o r}\cos\left(\frac{\Delta\nu}{2}\right)\right\}^3}$$

```
m=mu*t^2/((2*sqrt(norm(r_int)*norm(r_TGT))*cos((nu1-nu)/2))^3);
l=(norm(r_int)+norm(r_TGT))/(4*sqrt(norm(r_int)*norm(r_TGT))...
*cos((nu1-nu)/2))-1/2;
```

  - Find y

Escobal ([1965] 1985:192–194) shows how to develop the *second equation of Gauss*:

$$y^3 - y^2 = m x_2 = y^2(y-1) \qquad x_2 = \frac{\Delta E - \mathrm{SIN}(\Delta E)}{\mathrm{SIN}^3\left(\frac{\Delta E}{2}\right)}$$

HW 5

- Determine Transfer orbit using Lambert problem approach (Lambert-Gauss)
  - Find y

```
q0=2*atan(sqrt((1-norm(e))/1+norm(e))*tan(nu/2));
q=2*atan(sqrt((1-norm(e))/1+norm(e))*tan(nu1/2));
X=(q-q0-sin(q-q0))/sin((q-q0)/2)^3;
Y=[1, -1, 0, -m*X];
Y=roots(Y);
y=Y(3);
```

  - Find semiparameter of transfer orbit

```
p=y^2*norm(r_int)^2*norm(r_TGT)^2*sin(nu1-nu)^2/(mu*t^2);
```

HW 5

- Find delta v
  - Using f & g function

$$f = 1 - \frac{r}{p}(1 - \cos(\Delta\nu)) \qquad g = \frac{r r_o \sin(\Delta\nu)}{\sqrt{\mu p}}$$

$$\dot{f} = \sqrt{\frac{1}{p}} \tan\left(\frac{(\Delta\nu)}{2}\right)\left(\frac{1 - \cos(\Delta\nu)}{p} - \frac{1}{r} - \frac{1}{r_o}\right)$$

$$\dot{g} = 1 - \frac{r_o}{p}(1 - \cos(\Delta\nu))$$

$$\vec{v}_o = \frac{\vec{r} - f\vec{r}_o}{g} \qquad \vec{v} = \frac{\dot{g}\vec{r} - \vec{r}_o}{g}$$

```
f=1-r_TGT/p*(1-cos(nu1-nu));
g=norm(r_int)*norm(r_TGT)*sin(nu1-nu)/(sqrt(mu*p));
ff=sqrt(1/p)*tan((nu1-nu)/2)*((1-cos(nu1-nu))/p-1/norm(r_int)-1/norm(r_TGT));
gg=1-r_int/p*(1-cos(nu1-nu));
|
v_0=(r_TGT-f.*r_int)./g;
v=(gg.*r_TGT-r_int)./g;

delv_a=v_0-v_int;
delv_b=v_tgt-v;
```

HW 5

- Check for Earth impact

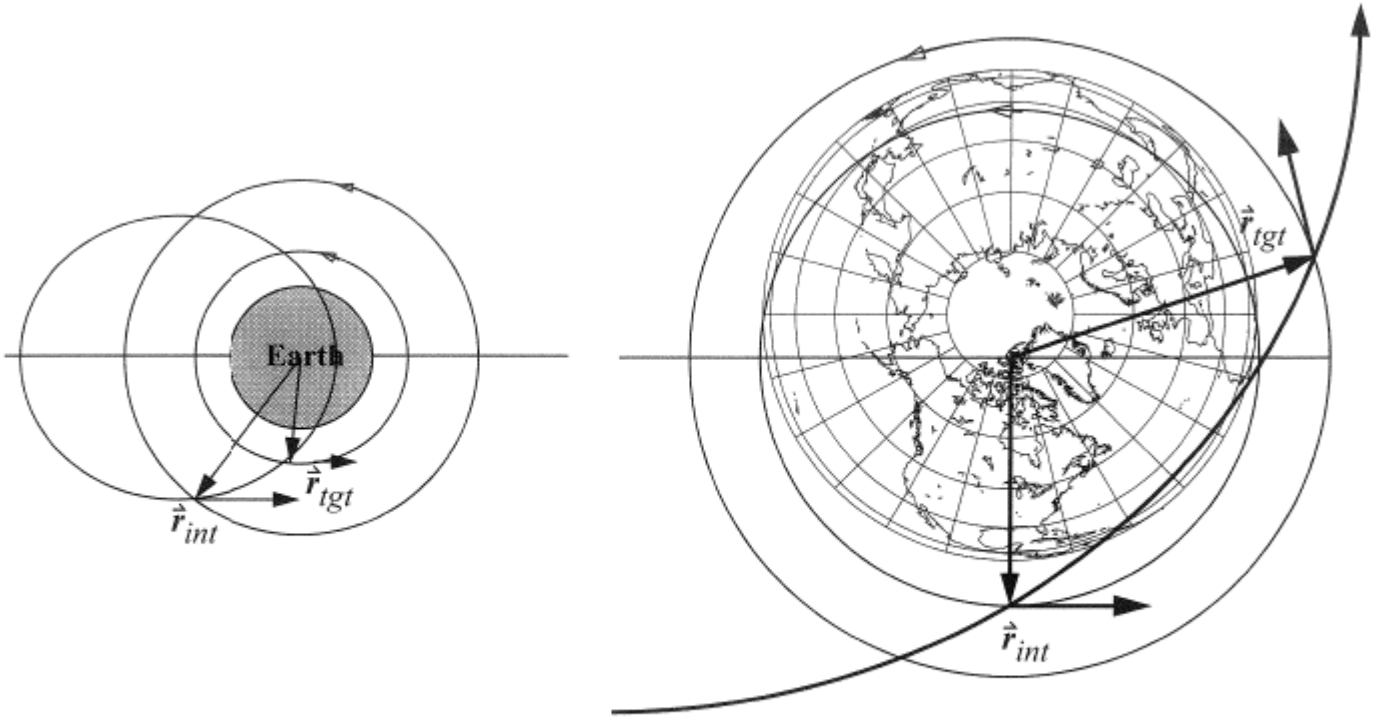

Figure 7-16. Geometry of Orbits that Can Hit the Earth. The orbit transfer on the left is valid because the transfer orbit intersects the Earth after the transfer. The transfer on the right would be unacceptable.

HW 5

- Check for Earth impact

ALGORITHM 59: ***HIT EARTH ALGORITHM 60:***

$$(\vec{r}_{int}, \vec{r}_{tgt}, \vec{v}_{trans_a}, \vec{v}_{trans_b}, a \Rightarrow r_p, \text{collision})$$

IF $((\vec{r}_{int} \cdot \vec{v}_{trans_a}) < 0.0)$ and $((\vec{r}_{tgt} \cdot \vec{v}_{trans_b}) > 0.0)$

$$\xi = \frac{v_{trans_a}^2}{2} - \frac{\mu}{r_{int}} = -\frac{\mu}{2a}$$

$$h_t = \left|\vec{r}_{int} \times \vec{v}_{trans_a}\right|$$

$$p = \frac{h_t^2}{\mu}$$

$$e = \sqrt{\frac{a-p}{a}}$$

$$r_p = a(1-e)$$

IF $r_p \le R_\oplus$ (1.0 ER, 6378.136 km)

Collision

ELSE

No Collision

HW 5

- Check for Earth impact

```
if dot(r_int,v_0)<0
    if dot(r_tgt,v)>0
    mu=398600;
    a=-mu/2*(norm(v_0)^2/2-mu/norm(r_int));
    h=cross(r_int,v_0);
    p=norm(h)^2/mu;
    e=sqrt((a-p)/a);
    r_p=a*(1-e);
    if r_p<=6378.136
        fprintf('Escape Earth₩n')
    else
        fprintf('We are safe₩n')
    end
    else
        fprintf('We are safe2₩n')
    end
else
fprintf('We are safe1₩n')
end
```

HW 6

HW 6

- main_hw4_5

```
xi = [6553;2674;3440;4.326;-1.925;-5.726];  % for HW4

opt = 3;    % 1:two-body / 2:j2,j3,j4 / 3:all perturbation

options = odeset('RelTol',1e-12,'AbsTol',1e-12);
[T Y] = ode45(@(t,x) hw4_5(t,x,opt),[0:10:86400],xi,options);

figure(1)
plot3(Y(:,1),Y(:,2),Y(:,3),'r'), grid on, hold on

figure(2)
subplot(3,2,1)
plot(T/86400,Y(:,1),'r'), grid on, hold on, legend('X Position')
subplot(3,2,3)
plot(T/86400,Y(:,2),'r'), grid on, hold on, legend('Y Position')
subplot(3,2,5)
plot(T/86400,Y(:,3),'r'), grid on, hold on, legend('Z Position')
subplot(3,2,2)
plot(T/86400,Y(:,4),'r'), grid on, hold on, legend('X Velocity')
subplot(3,2,4)
plot(T/86400,Y(:,5),'r'), grid on, hold on, legend('Y Velocity')
subplot(3,2,6)
plot(T/86400,Y(:,6),'r'), grid on, hold on, legend('Z Velocity')
```

HW 6

- Ode45
  - `varargout = ode45(ode,tspan,y0,options,varargin)`

```
options = odeset('RelTol',1e-12,'AbsTol',1e-12);
[T Y] = ode45(@(t,x) hw4_5(t,x,opt),[0:10:86400],xi,options);
```

HW 6

- Hw4_5
  - Setting initial condition

```
function Xdot = hw4_5(t,X,opt)

mu = 398600.4418;
Re = 6378.137;
we = [0;0;7.2921*10^-5];

r=sqrt(X(1)^2+X(2)^2+X(3)^2);
```

  - Case
    - ✓ Opt 1: two-body
    - ✓ Opt 2: zonal harmonics
    - ✓ Opt 3: zonal harmonics+air drag+3rd body (sun, moon)+solar radiation

HW 6

- Hw4_5
  - Opt 1: two-body

```
%% Case
if opt==1        % Two body case
    a = zeros(3,1);
```

  - Opt 2: zonal harmonics

	Moon	Mercury	Venus	Earth	Mars
Symbol	☾	☿	♀	⊕	♂
J_2	0.000 202 7	0.000 06	0.000 027	0.001 082 626 9	0.001 964
J_3				−0.000 002 532 3	0.000 036
J_4				−0.000 001 620 4	

HW 6

- Hw4_5
  - Opt 2: zonal harmonics

```
elseif opt==2   % J2, J3, J4 case
    %% Zonal harmonicss
    j2 = 1082.645*10^-6;
    j3 = -2.546*10^-6;
    j4 = -1.649*10^-6;

    a_j2=[-3*j2*mu*Re^2*X(1)/(2*r^5) * (1 - 5*X(3)^2/r^2) ;
        -3*j2*mu*Re^2*X(2)/(2*r^5) * (1 - 5*X(3)^2/r^2) ;
        -3*j2*mu*Re^2*X(3)/(2*r^5) * (3 - 5*X(3)^2/r^2) ];

    a_j3=[-5*j3*mu*Re^3*X(1)/(2*r^7) * (3*X(3) - 7*X(3)^3/r^2) ;
        -5*j3*mu*Re^3*X(2)/(2*r^7) * (3*X(3) - 7*X(3)^3/r^2) ;
        -5*j3*mu*Re^3/(2*r^7) * (6*X(3)^2 - 7*X(3)^4/r^2 - 3*r^2/5) ];

    a_j4=[15*j4*mu*Re^4*X(1)/(8*r^7) * (1 - 14*X(3)^2/r^2 + 21*X(3)^4/r^4) ;
        15*j4*mu*Re^4*X(2)/(8*r^7) * (1 - 14*X(3)^2/r^2 + 21*X(3)^4/r^4) ;
        15*j4*mu*Re^4*X(3)/(8*r^7) * (5 - 70*X(3)^2/(3*r^2) + 21*X(3)^4/r^4) ];
    %%
    a = a_j2 + a_j3 + a_j4;
```

HW 6

- Hw4_5
  - Opt 3: zonal harmonics++air drag+3rd body (sun, moon)+solar radiation

```
%% Air drag
ApM = 0.01; % 0.01 m^2/kg
Cd = 2.2;  % Drag coeff
den = 1.585*10^-12*exp(-(r-Re-1000)/60.828);   % Exponential model at 450-500km
v_rel = X(4:6) - cross(we,X(1:3));
a_air = -1/2*Cd*ApM*den*norm(v_rel)*v_rel;
%% 3rd body (Sun)
mu_sun = 132712440018;
jd = jday(2014, 12, 2, 13, 20, 10);
r_sun = (sun(jd))'*149597870.700;
a_sun = -mu_sun/norm(r_sun)^3*(X(1:3) - 3*(X(1:3))'*r_sun/norm(r_sun)^2*r_sun);
%% 3rd body (Moon)
mu_moon = 4902.8;
r_moon = (moon(jd))'*Re;
a_moon = -mu_moon/norm(r_moon)^3*(X(1:3) - 3*(X(1:3))'*...
    r_moon/norm(r_moon)^2*r_moon - 15/2*((X(1:3))'*r_moon/norm(r_moon)^2)^2*r_moon);
```

HW 6

- Hw4_5
  - Opt 3: zonal harmonics++air drag+3rd body (sun, moon)+solar radiation

```
%% Solar radiation
psr = 4.57*10^-6;   % kg*km/sec^2*m^2
Cr = 1.5;    % reflectivity
a_sr = psr*Cr*ApM*(X(1:3)-r_sun)/norm(X(1:3)-r_sun);
 a = a_j2 + a_j3 + a_j4 + a_air + a_sun + a_moon + a_sr;
```

  - Equation of motion

```
%% EOM
Xdot = [X(4);
        X(5);
        X(6);
        -mu/r^3*X(1) + a(1);
        -mu/r^3*X(2) + a(2);
        -mu/r^3*X(3) + a(3)];
```

HW 6

- Extra function
  - Jday

```
function jd = jday(yr, mon, day, hr, min, sec)

        % ------------------------  implementation   ------------------
        jd = 367.0 * yr  ...
             - floor( (7 * (yr + floor( (mon + 9) / 12.0) ) ) * 0.25 )   ...
             + floor( 275 * mon / 9.0 ) ...
             + day + 1721013.5  ...
             + ( (sec/60.0 + min ) / 60.0 + hr ) / 24.0;
```

  - Moon

```
function [rmoon, rtasc,decl] = moon ( jd );
```

  - Sun

```
function [rsun,rtasc,decl] = sun ( jd );
```

Thank you for listening